ABHI THIAGARAJAN

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EDUCATION

The University of Texas at Austin — Bachelor of Science, Biomedical Engineering — May 2028
Minor: Programming and Computation
Overall GPA: 4.00/4.00

Relevant Coursework — Data Structures and Algorithms, Introduction to Computing (Biomedical & Electrical), Differential and Integral Calculus, Multivariable Calculus, Differential Equations, Engineering Physics I & II, Biomedical Design

EXPERIENCE

University of Texas at Dallas – *Research Intern*; Dallas, TX — May 2024 – May 2025
- Built Python-based backend services for high-volume image/video processing, developing modular APIs, batch-processing tools, and validation pipelines using FastAPI, NumPy, TensorFlow, and Scikit-learn.
- Increased system performance by 18% through GPU-accelerated workloads, hyperparameter optimization, multiprocessing, and Linux profiling (Nvidia-SMI, vectorization).
- Developed internal CLI tools and backend automation utilities that reduced end-to-end pipeline latency by 25% and improved reliability across distributed environments.
- Implemented interactive data dashboards using Plotly/Dash, enabling real-time visualization of system metrics and improving debugging efficiency for engineering teams.

360DigiTMG – *Full-Stack/Data Engineering Intern;* Dallas, TX — June 2023 – August 2025
- Built scalable Python/SQL APIs and ETL pipelines for 50K+ records, enabling efficient data processing and analytics workflows.
- Developed dashboards with Plotly, Dash, and Power BI for dynamic metrics exploration, reducing manual workflow by 40%.
- Implemented Flask-based predictive services with ML models, exposing inference endpoints and ensuring >90% accuracy.

Wolfram – *Research Intern*; Boston, MA — May 2024 – August 2024
- Engineered distributed backend systems across 250K+ nodes using GPU acceleration & parallel processing, cutting runtime by 42%.
- Developed Python/Wolfram data pipelines and visualization tools for real-time monitoring of spatiotemporal simulation behavior.
- Optimized multithreading, memory profiling, and parallel workflows, boosting throughput across large-scale compute tasks.

PROJECTS

AI-Driven Cardiovascular Diagnostics Platform — Full-Stack Developer — March 2025
- Built a full-stack platform with FastAPI backend and Plotly/Dash frontend for real-time ECG analysis and risk prediction.
- Deployed cloud workflows on AWS Lambda and API Gateway, reducing model inference latency by 37% and improving scalability.
- Designed interactive clinician dashboards to visualize ECG metrics and predictions, cutting diagnostic review time by 30%.

CarbonAI – Sustainability Optimization Platform — Full-Stack Developer — August 2024
- Developed serverless Python backend and Dash frontend to deliver real-time AI-driven carbon reduction insights for clients.
- Built microservice APIs for LLM inference, scoring, and recommendation generation, supporting simultaneous users efficiently.
- Deployed full-stack app on AWS cloud, enabling 15+ organizations to track, visualize, and reduce carbon footprint by 20%.

ACCOMPLISHMENTS

- GE Girls STEAM Lead – LEAD instructor for 100+ students Python, ENG design, and robotics — 2025
- Wolfram Scholarship Award — 2024, 2025
- Wolfram Emerging Leaders – Top 10% of research interns — 2024
- WWT Hackathon Winner – ML diagnostic imaging solution — 2024
- Sustainability Innovation Award – AI carbon management — 2024
- Girls Recode Hacks Winner — 2023

SKILLS

Technical Skills: Python, Java, SQL, C++, JavaScript, HTML/CSS, TensorFlow, Scikit-learn, Wolfram Language
Certifications: AWS CCP, Azure Fundamentals, Power BI, Git, REST APIs, Wolfram Cloud
Interests: Puzzles, Formula 1, Shark Tank, Kayaking, Food Truck Hopping, National Parks, Volleyball, Hackathons, Matcha
Work Eligibility: Eligible to work in the U.S. with no restrictions